MILITARY SPECIFICATION

Transmitter, Rate of Flow, Fuel; TRU-26/A
(300 to 10,000 PPH)

This specification has been approved by
the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of remote indicating, fuel, rate of flow transmitter requiring frequency control of the input voltage.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATION

Military

MIL-P-116	Preservation, Method of
MIL-H-3136	Hydrocarbon Fluid, Standard Test
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5624	Fuel, Aircraft Turbine and Jet Engine, Grades JP-3, JP-4 and JP-5
MIL-F-7024	Fluids, Calibrating, For Aircraft Fuel, Systems Components
MIL-I-7566	Indicator System, Fuel Flowmeter, General Specification for
MIL-P-7936	Parts and Equipments, Aeronautical, Preparation of Delivery

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MS3134	Connector, Electrical Receptacle, Solder-Mounted

(When requesting specifications, standards, drawings and publication refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania Attention: Code CDS.)

3. REQUIREMENTS

3.1 Qualification.- The transmitter furnished under this specification shall be a product which has been tested and has passed the qualification tests herein.

3.2 General Specification.- The requirements of MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and Construction.-

3.3.1 General.- The transmitter shall be designed and calibrated to operate satisfactorily with a 300-5000-10000 pph indicator.

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3.3.1.1 The transmitter remote indication system shall operate as a synchro system in which the moving element consists of a 2-pole wound rotor, designed for operation with 115V 400 cycle single phase ac. The stator is 3 phase Y-connected. Frequency tolerances are shown in Figure 1.

3.3.2 Range.- The transmitter shall measure flow rated from 3000-5000-10000 pounds per hour.

3.3.3 Operating Pressure.- The high pressure transmitter shall operate under a pressure of 600 psi and withstand a pressure of 1600 psi without damage.

3.3.4 Pressure Drop.- The transmitter, by virtue of its design, shall not have appreciably more pressure drop under any fuel flow conditions when inoperative than when it is operating normally.

3.3.5 Temperature Range.- The transmitter shall be designed to withstand an ambient operating temperature range of -55° to +121°C and high temperature exposure to +150°C.

3.3.6 Electrical Receptacle.- The transmitter's electrical receptacles shall be in accordance with MS3134 and of the type shown in figure 1.

3.3.7 Power Consumption.- The power consumption of the transmitter shall be a maximum of 15 watts.

3.3.8 Dimensions.- The outline dimensions of the transmitter shall conform to figure 2.

3.3.9 Weight.- The weight of the complete fuel flowmeter transmitter shall not exceed 6.5 pounds.

3.3.10 Mounting.- Mounting bosses and spacing thereof shall be as specified on figure 2.

3.3.11 Flow Direction.- The direction of fuel flow through the transmitter shall be permanently and legibly marked by arrow as shown on figure 2 on both front and back.

3.3.12 Identification of Product.- Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Sampling, inspection and testing of the transmitters shall be as specified in MIL-I-7505, with the exceptions and additions as specified herein.

4.1.1 Applicable tolerances shall be as specified in section 3, and as specified in the following tables I, II and III.

4.1.2 Test Position.- Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.1.3 Test Fluid and Fuel.- The fluid used for calibrating and testing the transmitter shall be fluid conforming to Specification MIL-H-3136 Type I and MIL-F-7024 Type II or aviation fuel conforming to Specification MIL-F-5624 Grades JP-4 and JP-5.

4.1.4 Fuel Temperature.- The temperature of the test fuel during high temperature test runs shall be 38 ± 2°C. The fuel shall be pumped through the transmitter at the rate of 1000 pph for the last 30 minutes prior to scale and friction error calibration tests.

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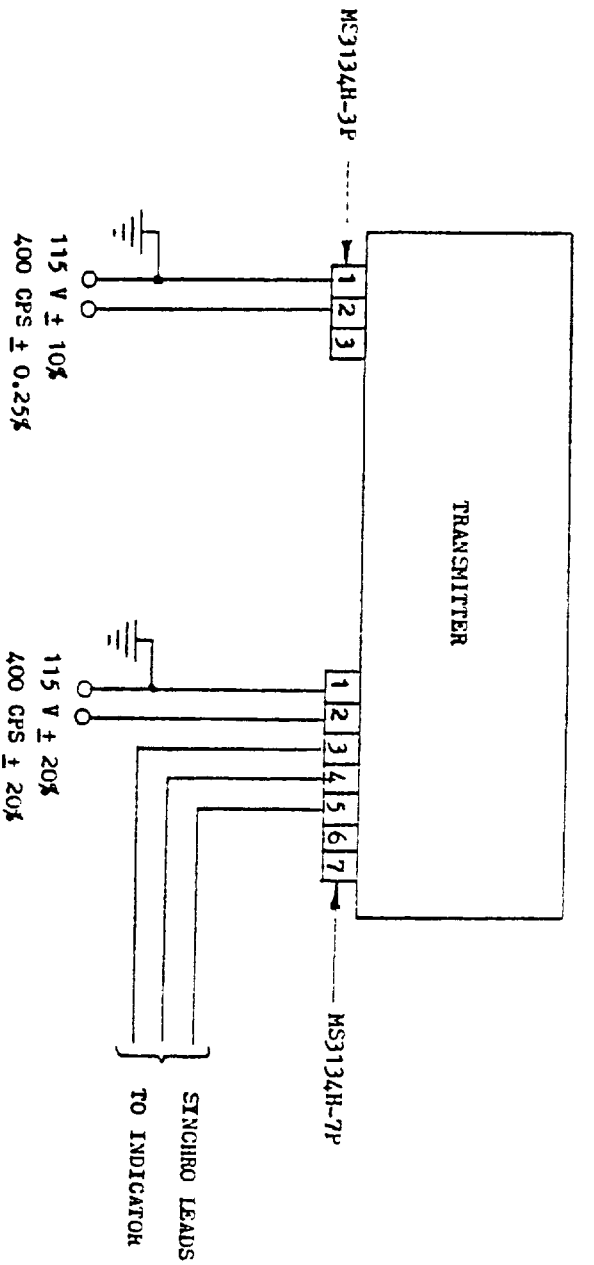
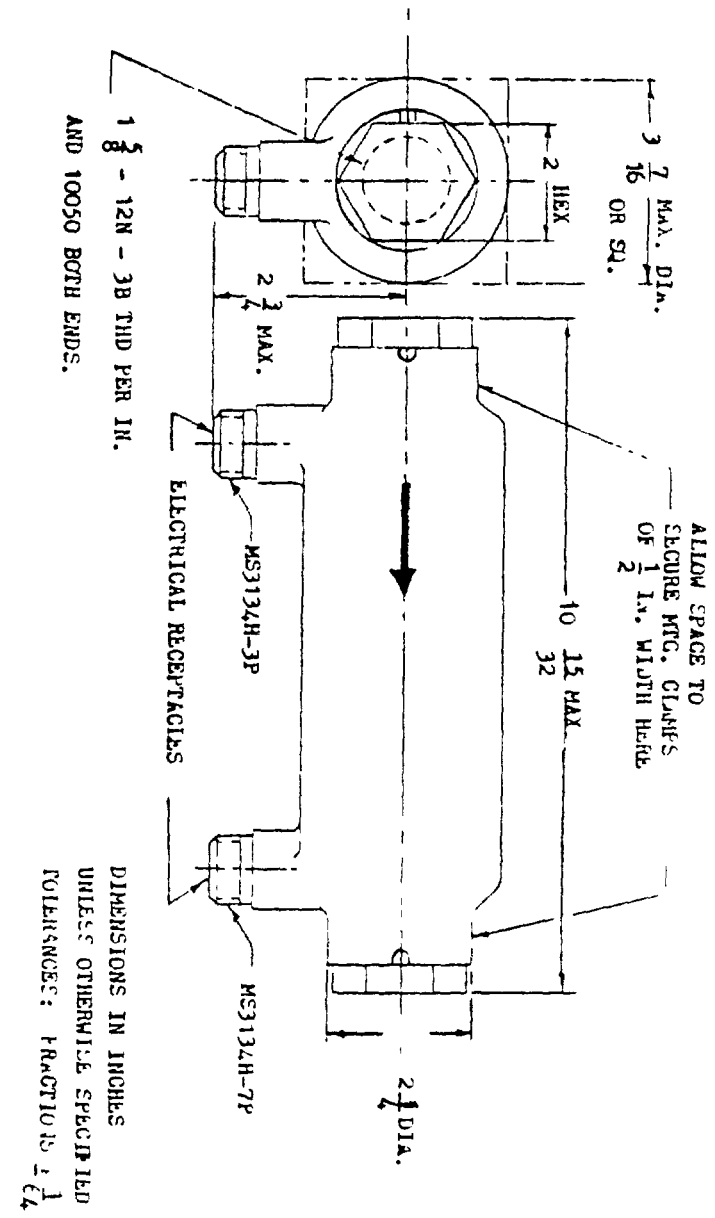


FIGURE 1

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TRANSMITTER

FIGURE 2

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TABLE I

Fuel Flowmeter Transmitter

Flow Rate (PPH)	Scale Error Tolerance (± PPH) Room Temp.	High Temp.	Low Temp.
300	45	55	75
500	35	45	65
1500	35	45	60
2500	30	40	55
3500	30	40	55
4500	45	60	85
6000	120	240	300
9000	170	340	425
10000	185	370	455

TABLE II

Test	Tolerance PPH ± 300-5000 PPH Scale	5000-10000 PPH Scale
Position Error	25	125
Friction		
Room Temperature	15	150
High Temperature	15	150
Low Temperature	20	200
Vibration		
Pointer Oscillation	50	250
Pointer Variation	50	250
Upscale-downscale reading variation	25	200
Voltage & Frequency Variation	Direct ratio to % variation of frequency	

TABLE III

Flow Rate PPH	Maximum Allowable Pressure Drop PSI
000	0.21

4.2 Qualification Tests.- The following tests specified in sampling plan B of MIL-I-7566 shall be performed for the qualification of the transmitter only:

Salt Spray
Radio Noise
Transmitter Substitute Fuel.

4.2.1 Additions and exceptions to the qualification testing requirements of MIL-I-7566 are as follows:

4.2.1.1 High Temperature Exposure.- The transmitter shall be subjected to a temperature of +150° ± 2°C for a period of 1 hour. The transmitter shall be in the inoperative condition with no fuel flowing through the instrument. At the end of the 1 hour exposure the transmitter shall be maintained at room temperature for not less than 4 hours. At the end of the 4 hour period the transmitter shall be tested for room temperature scale errors and friction. The tolerances as specified in tables I and II shall apply.

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4.2.1.2 Endurance.- The scale errors after endurance test shall not exceed the tolerances specified in table I for the room temperature scale error test by more than 40 pph in the lower scale range, and 100 pph in the upper range. Friction errors shall be within the tolerances specified in table II.

4.2.1.2.1 Endurance Duration.- In the event the elapsed time of the endurance test is less than 1200 hours when the specified number of flow applications has been completed, then the system shall be operated at 80 ± 5 percent of the range of the system until a total of 1200 hours of operation has been completed.

4.2.1.3 Contaminated Fuel Operation.- Test fuel containing dirt to the extent of 80 grams of foreign matter to 1000 gallons of fuel shall be circulated through the transmitter at approximately 2000 pph for one hour. A scale and friction error test shall then be made with the contaminated fuel and operation shall be satisfactory and within the room temperature tolerances. This foreign matter shall be considered to consist of not less than 68 percent SiO_2 and shall have a particle-size analysis as shown in table IV.

TABLE IV

Foreign Particle Size, Microns	Percent of Total Weight
0 - 5	39 ± 2
5 - 10	18 ± 3
10 - 20	16 ± 3
20 - 40	18 ± 3
over 40	9 ± 3
Through a 200-mesh screen	100 percent

4.2.1.4 Response Time.- The fuel flow shall be changed from 0 to 6000 pph by means of a rapid action cut-off valve in the fuel line. The time for the indication of a master indicator connected to the transmitter to come within one percent of its steady state position shall be within 5 seconds.

4.2.1.5 Shock Test.- The test shall be in accordance with procedure I of MIL-E-5272. The transmitter shall be subjected to impact shocks of 15g, each shock impulse having a time duration 7 ± 2 milliseconds. Three shocks shall be applied in each of the following directions:

a. Along the longitudinal axis
b. At 45° to the longitudinal axis in one direction.

4.3 Sampling Test.-

4.3.1 Sampling Plan A.-

4.3.1.1 Pressure.- The pressure drops of the transmitter shall be determined for the rates of flow specified and shall not exceed the tolerance specified in Table III.

4.3.1.2 Transmitter Vibration During Pointer Oscillation and Vibration Tests.- The pointer oscillation and variation test errors shall be determined when the transmitter is subjected to the frequency range at vibratory accelerations or double amplitudes shown below:

Frequency	Amplitude
5 - 100 cps	0.010 inch double amplitude
100 - 500 cps	5 g acceleration

4.3.1.3 Frequency Variation.- The error, in percent of indicated flow, shall be equal to the percentage change in the supply frequency from the normal 400 cps.

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4.3.2 Sampling Plan B.-

4.3.2.1 Altitude Test .- This test shall not be required.

4.3.2.2 Low Temperature Operation.- The transmitter shall be subjected to a temperature of -65° ± 2°C for a period of not less than 24 hours, and then operated at this temperature over the scale, temperature shall then be raised to -55° ± 2°C for a period of not less than 24 hours. The scale errors and friction shall not exceed the low temperature tolerances as shown in tables I and II. The transmitter shall be subjected to the room temperature scale error test not less than 1 hour following the cold exposure test. The room temperature tolerances as specified in tables I and II shall apply.

4.3.2.3 High Temperature Operation.- The transmitter shall be subjected to an ambient temperature of 121°C for a period of 4 hours with fuel in the instrument, but not flowing. During the last 30 minutes of this period, fuel at 38° ± 2°C shall be pumped through the transmitter at a rate of 1,000 pounds per hour. The transmitter shall then be tested for scale errors and friction under the same conditions. The scale errors and friction shall not exceed the tolerances specified in tables I and II.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- The transmitter shall be packaged in accordance with MIL-P-7936. The method of preservation shall be MIL-P-116, Method 1A5 omitting preservation compound, using metal reusable containers. (See 6.2).

5.1.2 Level C.- The transmitter shall be packaged in accordance with standard commercial practices.

5.2 Packing.-

5.2.1 Levels A and B.- The transmitter shall be packed in accordance with MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract. (See 6.2).

5.2.2 Level C.- The transmitter shall be packed in accordance with standard commercial practices.

5.3 Marking.- The interior and exterior containers shall be marked as specified in MIL-P-7936.

5.3.1 Precautionary Marking.- The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

6. NOTES

6.1 Intended Use.- The type fuel flowmeter transmitter covered by this specification is intended to indicate the rate of flow of fuel in pounds per hour being consumed by an aircraft jet engine when used in conjunction with a suitable indicator.

6.2 Ordering Data.- Procurement documents should specify the level of packaging and packing desired.

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6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by the specification may be obtained from the Bureau of Aeronautics, Department of the Navy, Washington 25, D. C.

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